

Punjab University College of Information Technology (PUCIT)

University of the Punjab, Lahore

Course: Programming Fundamentals (Final Examination)

Time Allowed: 120 Minutes | **Total Marks:** 160

Degree: (Encircle) BSE / BCS / BIT | **Section:** (Encircle) Morning / Afternoon

INSTRUCTIONS

1. **Do Not Start Until Advised:** Pray before beginning; stay relaxed and honest.
 2. **Closed Book:** No notes, calculators, or mobile devices are permitted.
 3. **No Queries:** Understanding the technical requirements of the question is part of the assessment.
 4. **Formatting:** If the space provided is insufficient, utilize two columns or the extra page.
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Question #01 (CLO-1) [4 parts, 5 each | Total 20]

Answer the following conceptual questions precisely in your own words.

- a. Define **Memory Alignment** in C++ structures. Why might a `struct` with one `int` and one `char` occupy 8 bytes instead of 5?
 - b. Explain the difference between `seekg` and `seekp`. Under what specific scenario would you use `ios::cur`?
 - c. Differentiate between **Dynamic Memory** and **Static Memory**. Which one is managed via the 'Heap', and how is it deallocated?
 - d. What is a **Null Terminator** in C-style strings? Explain what happens if a `char` array is printed using `cout` but lacks the `\0` character.
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Question #02 (CLO-2) [5 parts, 4 each | Total 20]

*Identify and correct logical/syntax errors, then provide the **Exact Output**.*

Code Segment	Corrections	Exact Output
A. <code>int a[3]={5,10,15}; int *p=a;</code> <code>cout << *(p++) << " " << *p;</code>		
B. <code>struct Box { int l, w; };</code> <code>Box b; b->l = 10; cout << b.l;</code>		
C. <code>char s[] = "PUCIT";</code> <code>for(int i=0; s[i]; i++) s[i]+=1;</code> <code>cout << s;</code>		
D. <code>int x=5, y=2;</code> <code>float z = x / y; cout << fixed << z;</code>		
E. <code>ofstream f("d.dat", ios::app);</code> <code>f.seekp(0); f << "A"; f.close();</code>		

Question #03 (CLO-2) [4 parts, 5 each | Total 20]

Write short C++ code segments for the following tasks.

a. Given `int mat[4][4]`, write a loop to calculate the product of all elements in the **secondary diagonal**.

b. Define a `struct Complex` with `real` and `imag` (`float`). Write a function `Complex conjugate(Complex c)` that returns the conjugate.

c. Write a code segment that uses `ifstream` to count the total number of lines in a text file named `data.txt`.

d. Create an `enum Semester { FALL, SPRING, SUMMER }`. Write a `switch` statement that prints the name of the semester based on an `enum` variable `s`.

Question #04 (CLO-2) [2 parts, 20 each | Total 40]

Implement the following complete functions.

a. **File Handling (Pakistani Cities):** Referencing your `pakcities.txt` format (Name, Lat, Lon), write a function `void listWesternCities(string fileName, double lonLimit)`. This function must read the file and display only the names of cities whose **Longitude** is less than `lonLimit`. Handle the file stream carefully using `db.getline()` and `db.ignore()` as seen in `pakcities.cpp`.

b. **2D Array Analysis (Temperature Data):** Given a 2D array `float temp[3][7]` (3 shifts, 7 days), write a function `void reportExtremes(float temp[3][7])`. The function must find and print:

1. The **hottest shift** (the row index with the highest average temperature).
 2. The **overall minimum temperature** recorded during the entire week and its coordinates (row, col).
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Question #05 (CLO-3) [6 parts, 10 each | Total 60]

Advanced Analytical Problems.

a. **Polynomial Subtraction:** Using the `struct Polynomial { int degree; unsigned int coefficients[21]; }`, write a function `Polynomial subtract(const Polynomial &p1, const Polynomial &p2)`. Ensure you correctly handle cases where the degrees are different.

b. **Terminator-Based Arrays:** Write a function `void findPeaks(int data[])` that accepts an array of positive integers ending with `-1`. A "peak" is an element greater than its immediate neighbors. Print all peaks found.

c. Random Access Update: You have a binary file `grades.dat` containing records of `struct Record { int id; float score; }`. Write a function `void updateScore(int targetID, float newScore)` that finds the record by ID and updates only that specific score using `seekp`.

d. Dynamic Memory (Jagged Array): Write a code segment to dynamically allocate a 2D `double` array where the number of rows is `R`, but each row `i` has a different number of columns stored in an array `int colSizes[R]`. Include the deallocation code.

e. C-String Comparison: Write a function `int compareAlpha(char s1[], char s2[])` that performs a case-insensitive comparison of two C-strings. Return 0 if equal, 1 if `s1 > s2`, and -1 otherwise. **Do not use `string` class.**

f. Matrix Saddle Point: Write a function for a `5x5` matrix to find a "Saddle Point": an element which is the **minimum** in its row and the **maximum** in its column.

HONOUR CODE: In the movie *Rise of the Planet of the Apes*, Caesar is seen solving the **Lucas Tower** (Tower of Hanoi). What is the minimum number of moves to solve this puzzle for 3 disks?

SIGNATURE: _____ **ROLL NO:** _____